

C Programming

❖ Control Instruction

- A control instruction is a statement in a program that controls the flow of execution of the program.

It decides:

- ✓ Which statement will execute
- ✓ When it will execute
- ✓ How many times it will execute

Types of Control Instruction

1. Decision / Statement Control Instruction (if,if-else,switch)
2. Iterative / Loop Control Instruction (for, while, do-while)
3. Jump Control Instruction (break, continue, goto, return)

1. Decision Control Instruction / Selection Control Instruction

- a. If (check only 1 condition)
- b. if else (check only 2 condition)
- c. else if (check 2 or more than 2 condition)
- d. ?: (Conditional Operator/Ternary Operator)
- e. switch

a. if

Syntax :→ if (Condition)
 {

```

        -----;
        -----;
    }

```

Q. Write a program in c take any number and check is odd or even ?.

```

#include<stdio.h>
Int main()
{
    Int a;
    Printf("Enter any number = ");
    Scanf("%d",&a);
    If(a%2==0)
    {
        printf (" this is Even number");
    }
    If(a%2==1)
    {
        Printf("This is Odd Number");
    }
    Return 0;
}

```

b. if else

Syntax :→

```

    if (Condition)
    {
        -----;
        -----;
    }
    Else
    {

```

```
        -----;  
        -----;  
    }
```

Q. Write a program in c take any number and check is odd or even ?.

```
#include<stdio.h>  
Int main()  
{  
    Int a;  
    Printf("Enter any number = ");  
    Scanf("%d",&a);  
    If(a%2==0)  
    {  
        printf (" this is Even number");  
    }  
    else  
    {  
        Printf("This is Odd Number");  
    }  
    Return 0;  
}
```

c. Nested if / Nested if else / else if

```
If(Condition)
```

```
{
```

```
    if(Condition)
```

```
    {
```

```
        -----;
```

```
        -----;
```

```
    }
```

```
}
```

```
if(Condition)
```

```
{
```

```
    if(Condition)
```

```
    {
```

```
        -----;
```

```
        -----;
```

```
    else
```

```
        -----;
```

```
        -----;
```

```
    }
```

```
}
```

```
if(Condition)
```

```
{
```

```
    if(Condition)
```

```
    {
```

```
        -----;
```

```
        -----;
```

```
    else
```

```
        -----;
```

```
        -----;
```

```
    }
```

```
    else
```

```
        if(Condition)
```

```
        {
```

```
            -----;
```

```
            -----;
```

```
        else
```

```
            -----;
```

```
            -----;
```

```
        }
```

```
    }
```

d. Ternary Operator

Used to condition.

Syntax :→

Condition ? Message1: Message2;

Example :→

a>0 ? Even: Odd;

Q. Write a program in c take any number and check is odd or even ?.

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
int main()
```

```
{
```

```
int a;  
printf("enter any number a=");  
scanf("%d",&a);  
a%2==0? printf("Even"): printf("odd"); OR  
printf(a%2==0? "even":"odd");  
getch();  
}
```

Output

enter any number a=6

Even

All Program written minimum two methods.

◆ Basic Level

1. Check whether a number is even or odd.
2. Check whether a number is positive, negative, or zero.
3. Check whether a number is divisible by 5 or not.

◆ Intermediate Level

4. Find the greater number between two numbers.
5. Find the largest number among three numbers.
6. Check whether a character is a vowel or consonant.
7. Check whether a year is a leap year or not.

◆ Advanced Level

8. Check whether a number is divisible by both 3 and 7.
9. Check whether a number is greater than 100 and less than 500.
10. Check whether a student is pass, fail, or distinction based on marks.

e. switch

- In C language, a switch statement is a control statement used to make decisions based on the value of a variable or expression. It allows you to choose one block of code to execute from many options.

- Syntax

```
Switch (expression)
{
    Case constant/character:
        -----;
        -----;
    Case constant/character:
        -----;
        -----;
    Case constant/character:
        -----;
        -----;
    Default:
        -----;
        -----;
}
```

Q. WAP to print Week name.

```
#include<stdio.h> // printf() scanf()
#include<conio.h> // getch()
int main()
{
    int a;
    printf("Enter Week Number = ");
    scanf("%d",&a);
    switch (a)
    {
        case 1:
            {
                printf("Sunday");
                break;
            }
    }
```

```
}  
case 2:  
{  
    printf("Monday");  
    break;  
}  
case 3:  
{  
    printf("Tuesday");  
    break;  
}  
case 4:  
{  
    printf("Wednesday");  
    break;  
}  
case 5:  
{  
    printf("Thursday");  
    break;  
}  
case 6:  
{  
    printf("Friday");  
    break;  
}  
case 7:  
{  
    printf("Saturday");  
    break;  
}  
default:  
{  
    printf("Invalid key");  
    break;  
}
```

```
}
```

```
    getch();  
    return 0;
```

```
}
```

- 1. Define a game like program, in which user has to enter an even number to win the game. User will get at most 3 chances.**
- 2. WAP to print Months name.**
- 3. WAP to make a calculator (addition, subtraction, multiplication, division, odd-even, percentage)**
- 4. WAP to print 1st, 2nd, 3rd and fail .**